

Project Title: Statistical Analysis of the Magnificent 5 within the US Equity Market

Programming for Data Analytics (HDSADINTJAN26I)

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1 Introduction

Nowadays, the financial markets create vast amounts of data every day. That is the reason that systematic programming analysis is needed in asset management, equity research, and regulatory oversight. This project focuses on the exploration of the Magnificent Five companies, which are Apple, Microsoft, Alphabet, Amazon, and Meta. These are US technology companies, and together account for a disproportionate share of the S&P 500 (Investopedia, 2024). This project aims to run some statistical tests that can be presented by placing them inside the wider context of U.S.-listed companies on the NASDAQ exchange.

This analysis project combines two datasets: a five-year time series of daily prices and volume data for the five magnificent companies, and a cross-sectional dataset of around seven thousand NASDAQ-listed companies. This project will present five different insights, such as descriptive analysis, chi-square test, one-way ANOVA, and Welch's test, which will reveal interesting answers about how these companies relate to the equity market.

2 Research Objectives and Questions

In this project, the goal will be to apply various methods of statistical analysis to find out how exactly the Magnificent fit into the broader universe of the US stock market, and what makes them unique, as well as what makes them similar to other sectors. The following are the research questions that will be answered in the following contents:

1. What does the combined performance and company profile of the Magnificent 5 look like? (Insight 1)
2. Are certain sectors significantly more likely to contain larger-market-cap companies than others? (Insight 2)
3. Do the five Magnificent 5 stocks have statistically different mean daily returns? (Insight 3)

4. Do Technology companies trade more actively than Financial companies on NASDAQ? (Insight 4)

3 Dataset Collection & Description

Dataset 1 (Magnificent5_stockprice.csv) consists of a time series dataset containing historical information about daily prices for the open, high, low, close and volume for all five companies for the last five years. The data was obtained from Yahoo Finance (Yahoo Finance, 2021) via the open source yfinance Python library (Aroussi, 2017) and comprises approximately 6,275 rows over seven columns.

```
#Convert Date from string to datetime type

df_price["Date"] = pd.to_datetime(df_price["Date"])
df_price.info()

✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6280 entries, 0 to 6279
Data columns (total 7 columns):
#   Column  Non-Null Count  Dtype  
---  -
0   Date    6280 non-null   datetime64[ns]
1   Ticker  6280 non-null   object  
2   Open    6280 non-null   float64 
3   High    6280 non-null   float64 
4   Low     6280 non-null   float64 
5   Close   6280 non-null   float64 
6   Volume  6280 non-null   int64   
dtypes: datetime64[ns](1), float64(4), int64(1), object(1)
memory usage: 343.6+ KB
```

Dataset 2 (nasdaq_screener_1777050489651.csv) consists of cross-sectional data that contains information about all sectors currently traded on the NASDAQ market, obtained through the NASDAQ stock screener (NASDAQ, 2024). This dataset has 7,047 rows and 11 columns.

```
df_nasdaq.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7047 entries, 0 to 7046
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Symbol          7046 non-null   object
1   Name            7047 non-null   object
2   Last Sale       7047 non-null   float64
3   Net Change      7047 non-null   float64
4   % Change        7047 non-null   float64
5   Market Cap      6649 non-null   float64
6   Country         6780 non-null   object
7   IPO Year        4030 non-null   float64
8   Volume          7047 non-null   int64
9   Sector          6311 non-null   object
10  Industry        6310 non-null   object
dtypes: float64(5), int64(1), object(5)
memory usage: 605.7+ KB
```

These two datasets are purposely made complementary to each other rather than similar in order to provide different perspectives to the magnificent five stocks.

4 Dataset Preprocessing

The first Dataset (Magnificent5_stockprice.csv) was constructed by looping through the five tickers and downloading each one with `yf.download (auto_adjust=True)`. The returned Yahoo Finance DataFrame has a multi-index on its column axis, which is flattened to a single-index format for easier data processing. A column named Ticker is appended to each downloaded DataFrame, then the five DataFrames are joined into a single-long DataFrame with `pd.concat`. Finally, this is exported as the Magnificent5_stockprice_clean.csv, sorted first by Date and then by Ticker. No missing data and no duplicates were present in the raw dataset.

```
# Check for missing values and duplicate rows

print(df_price.isna().sum())
print(df_price.isna().any(axis=1).sum())
print(df_price.duplicated().sum())
```

✓ 0.0s

```
Date      0
Ticker    0
Open      0
High      0
Low       0
Close     0
Volume    0
dtype: int64
0
0
```

The second dataset (nasdaq_screener_1777050489651.csv) also required more cleaning. There were two variables in the text form: the Last Sale, such as “\$116.12”, and the % Change, like “1.486%”, to only numeric. Using str.replace, the symbols were deleted and turned both variables into floats. It was also deleted a row where there was no symbol information using the function dropna(subset=['Symbol']). All missing data in the Market Cap (398), Country (267), IPO Year (3,017), Sector (736), and Industry (737) remained in the original data frame. In the data analysis part, each insight will use only those variables necessary for its calculations. Therefore, it is not necessary to delete all missing values but rather those used by the insight itself.

```
#Check for missing values and duplicate rows
print(df_nasdaq.isna().sum())
print(df_nasdaq.isna().any(axis=1).sum())
print(df_nasdaq.duplicated().sum())
```

✓ 0.0s

```
Symbol      1
Name        0
Last Sale   0
Net Change  0
% Change    0
Market Cap  398
Country     267
IPO Year    3017
Volume      0
Sector      736
Industry    737
dtype: int64
3507
0
```

5 Dataset Exploration

After the data cleaning process, the cleaned Dataset 1 (Magnificent5_stockprice_clean.csv) has about 6275 days of OHLCV data for the five

tickers, with no missing entries, no duplicate rows, and the date range covers the last five years. The closing prices vary between the five companies, and the trading volumes are heavily right-skewed.

```
# Quick range check
print(df_price["Close"].min())
print(df_price["Close"].max())
print(df_price["Volume"].min())
print(df_price["Volume"].max())
```

✓ 0.0s

81.81999969482422
788.1489868164062
4726100
318679900

The cleaned dataset 2 (NASDAQ_screener_clean.csv) includes 7,046 companies in eleven sectors and a Miscellaneous category. The Finance sector accounts for the largest number of companies (1,573), followed by Health Care (1,093) and Consumer Discretionary (1,029), while Miscellaneous has the lowest number of companies (63). The technology sector contains 770 companies that make up the Magnificent 5 group. In terms of market capitalization, the smallest companies are worth less than one million dollars, whereas the biggest companies are worth several trillion.

```
df_nasdaq.describe()
```

✓ 0.0s

	Last Sale	Net Change	% Change	Market Cap	IPO Year	Volume
count	7047.000000	7047.000000	7047.000000	6.649000e+03	4030.000000	7.047000e+03
mean	146.263224	-0.347754	0.133483	1.485410e+10	2015.616377	1.153512e+06
std	8378.914870	34.278173	5.358675	1.343026e+11	10.998081	2.538828e+07
min	0.001500	-2865.000000	-93.300000	0.000000e+00	1925.000000	1.000000e+00
25%	3.953300	-0.080000	-0.920500	2.786747e+07	2012.000000	1.070800e+04
50%	14.035000	0.000000	0.000000	4.906272e+08	2020.000000	8.021900e+04
75%	35.220000	0.095000	0.977500	3.795452e+09	2024.000000	3.712385e+05
max	703300.000000	116.570000	174.503000	5.108591e+12	2026.000000	2.091383e+09

6 Data Analysis & Insights

6.1 Insight 1- Profile of the Magnificent 5

Question: What does the combined performance and company profile of the Magnificent 5 look like?

To illustrate this insight, the time-series statistics were calculated and extracted from the 'Magnificent5_stockprice' dataset, and the companies' attributes were extracted from the 'NASDAQ_screener' dataset. Loading the clean version of both files with 'pd.read_csv' by using a defined function with friendly error messages. After that, Figure 1 was created with the cumulative return value as each stock normalized to 1\$.

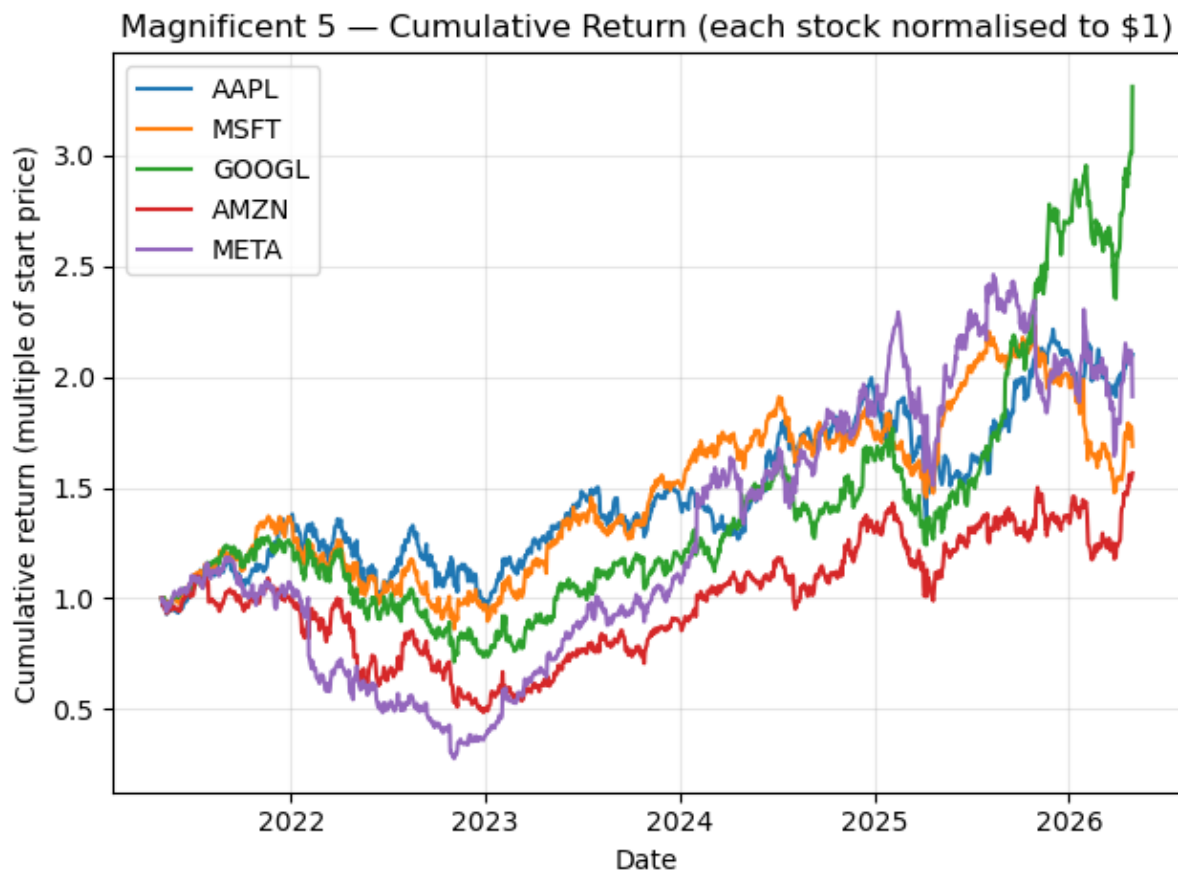


Figure 1: Cumulative return of Magnificent 5

Three metrics were calculated for each ticker from the daily closing prices using small functions. They are the total return over the five-year window, average daily return by using 'pct_change().mean()', and annualized volatility computed by the standard deviation of daily returns multiplied by the square root of 252 (the standard number of trading days per year). To describe the profiles of these Magnificent Five companies, the columns such as Sector, Industry, and Market Cap were pulled from the NASDAQ_screener and then merged with the performance statistics into the five-row table.

Magnificent 5 — Combined Profile Table

Ticker	Sector	Industry	Market Cap (\$B)	Total Return %	Average Daily Return %	Cumulative Annualized Volatility %
AAPL	Technology	Computer Manufacturing	3969	119.2	0.08	27.5
MSFT	Technology	Computer Software: Prepackaged Software	3114	71.1	0.06	26.4
GOOGL	Technology	Computer Software: Programming Data	4156	230.7	0.11	31.2
AMZN	Consumer Discretionary	Catalog/Specialty Distribution	2816	58.6	0.06	35.5
META	Technology	Computer Software: Programming Data	1715	91.1	0.09	43.9

Figure 2: The Magnificent 5- profile table

The results show the meaningful variation across the company. As shown in Figure 2, Google led with a total return of 230.7%, followed by Meta (119.2%) and Apple (91.1%); Microsoft (71.1%) and Amazon (58.6%) trailed. On the other hand, volatility ranked differently. Meta was the most volatile company at 43.9% annualized, while Microsoft was the steadiest at 26.4%, making the second-lowest return. According to the market capitalization, Google was the largest at roughly \$4.16 trillion, with Meta the smallest at 1.72 trillion. In the sector column, only Amazon sat outside the Technology sector by being classified as Consumer Discretionary because of its retail online business.

Conclusion

Although they are marked as a single group, they differ greatly in return (58.6%–230.7%), volatility (26.4%–43.9%), and market capitalisation (\$1.7T–\$4.2T). They shouldn't really be treated as a single investment.

6.2 Insight 2 — Chi-Square Test of Independence: Sector × Market Cap Tier

Question: Are certain sectors significantly more likely to host larger-market-cap companies?

This insight tests about a company's sector and market capitalization tier are statistically independent on the Nasdaq stock exchange. After loading the cleaned screener dataset, the first step is to drop the rows of missing Sector or Market Capitalization values (last 6307 companies for analysis), and then each remaining company was assigned to one of four standard tiers (Seth, 2024), which are Mega-Cap, Large Cap, Mid-Cap, and Small-Cap, respectively, using a defined function. After that, a contingency table was created by using `pd.crosstab` method, and the test was run with `scipy.stats.chi2_contingency`.

The test resulted in a chi-square statistic of 416.86 (df=33, p-value <0.001). The resulting p-value is well below 0.05; therefore, the null hypothesis of independence is rejected, and it is statistically significant sector and market cap tier are dependent on. The value of Cramer's V is 0.148, which indicates a weak association strength between the variables.

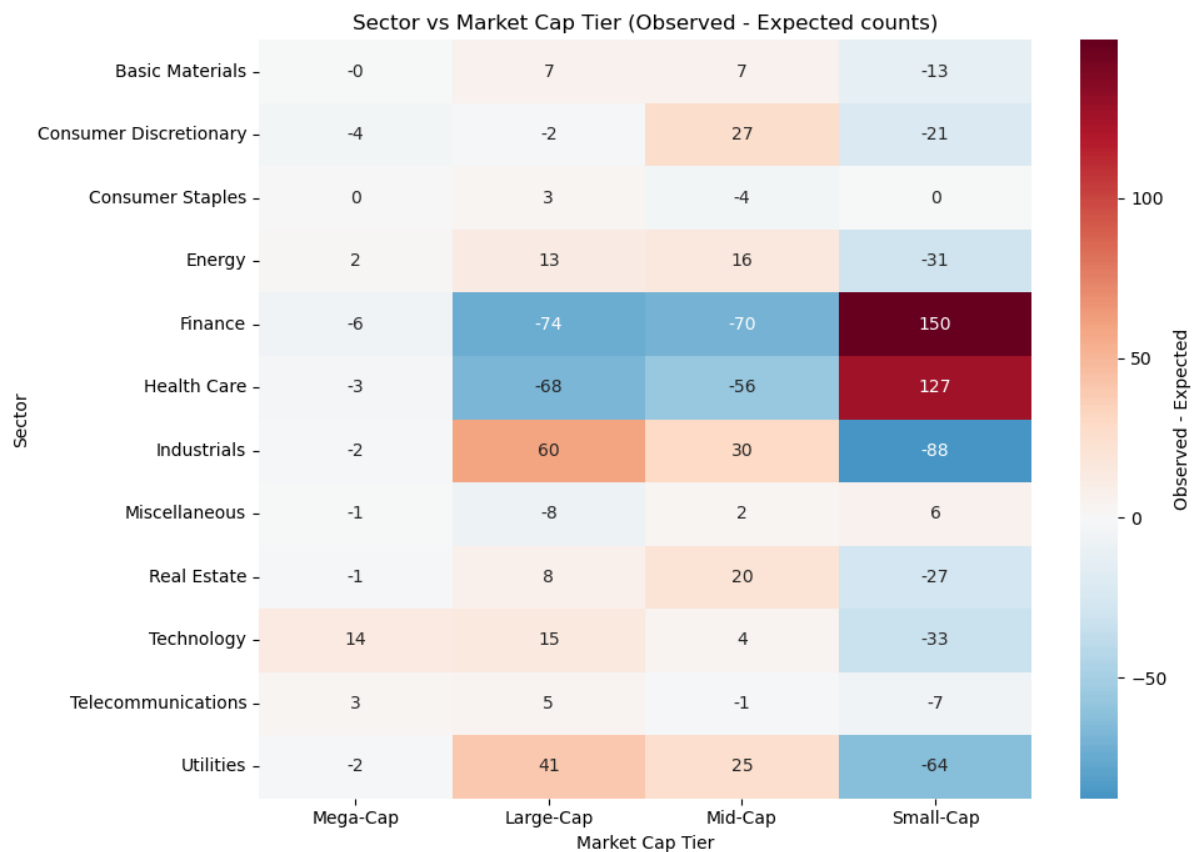


Figure 3: The residual heatmap of Sector Vs Market Cap Tier

The pattern shows very clearly in the residual heatmap (Figure 3). The technology sector is the only one with a positive Mega-Cap residual (+14), which supports the common retail trader belief that the largest companies concentrate on tech. Finance and Health Care are dramatically overrepresented in Small-Cap (+150 and +127, respectively). On the other hand, Industrials and Utilities were with strong Mid and Large concentrations and far fewer Small-Cap companies than expected.

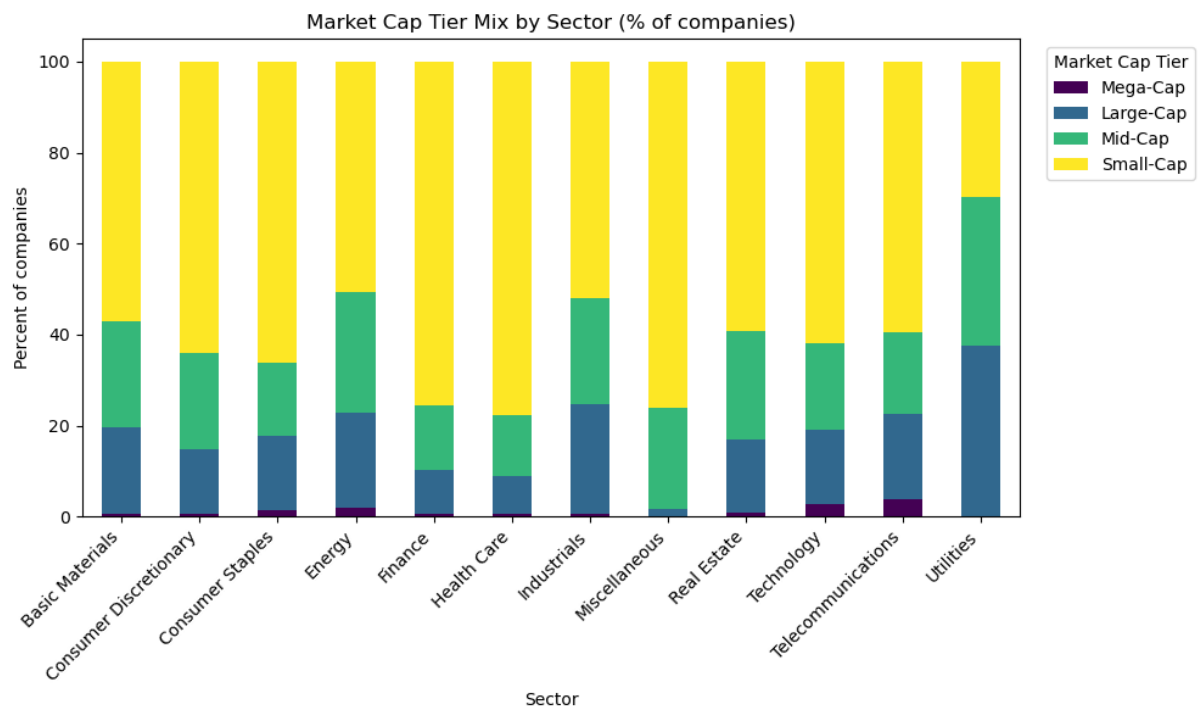


Figure 4: The stacked bar chart of Market Cap Tier Mix by sector

The stacked bar chart in Figure 4 can confirm visually that Utilities is the largest-tailed sector at around 70% Mid-Cap or above, while Finance and Health care are roughly 75% 80% Small-Cap.

Conclusion

To conclude insight 2, the common belief of “the giants are mostly Tech” is statistically supported, but the broader finding is how dramatically the market capitalization size distributions can differ by the sectors.

6.3 Insight 3 — One-Way ANOVA on Daily Returns by Company

Question: Do the Magnificent 5 stocks have statistically different mean daily returns?

This insight tests whether the magnificent five companies have statistically different daily return means over the five years. After loading the cleaned price data with the `read_csv` function, the daily returns were computed for each ticker by applying pandas such as `close.pct_change()` and `dropna()`. After that built a dictionary of daily return series per ticker to separate series because the one-way ANOVA test will compare them as five groups.

When printing the statistics summary, the means are very similar with MSFT 0.0566%, AMZN 0.0617%, AAPL 0.0775%, META 0.0904%, and GOOGL 0.1146%, respectively. The standard deviation, which is also called daily volatility, ranged from 1.66% to

2.76%. Each group included 1255 daily observations. The ANOVA test resulted in an F-statistic of 0.1545 and a p-value of 0.96, which is far above the 0.05 threshold. Therefore, the null hypothesis of equal means cannot be rejected. We can conclude that the five stocks have statistically equivalent mean daily returns. The boxplot (Figure 5) represents almost the same medians of the daily return of each ticker.

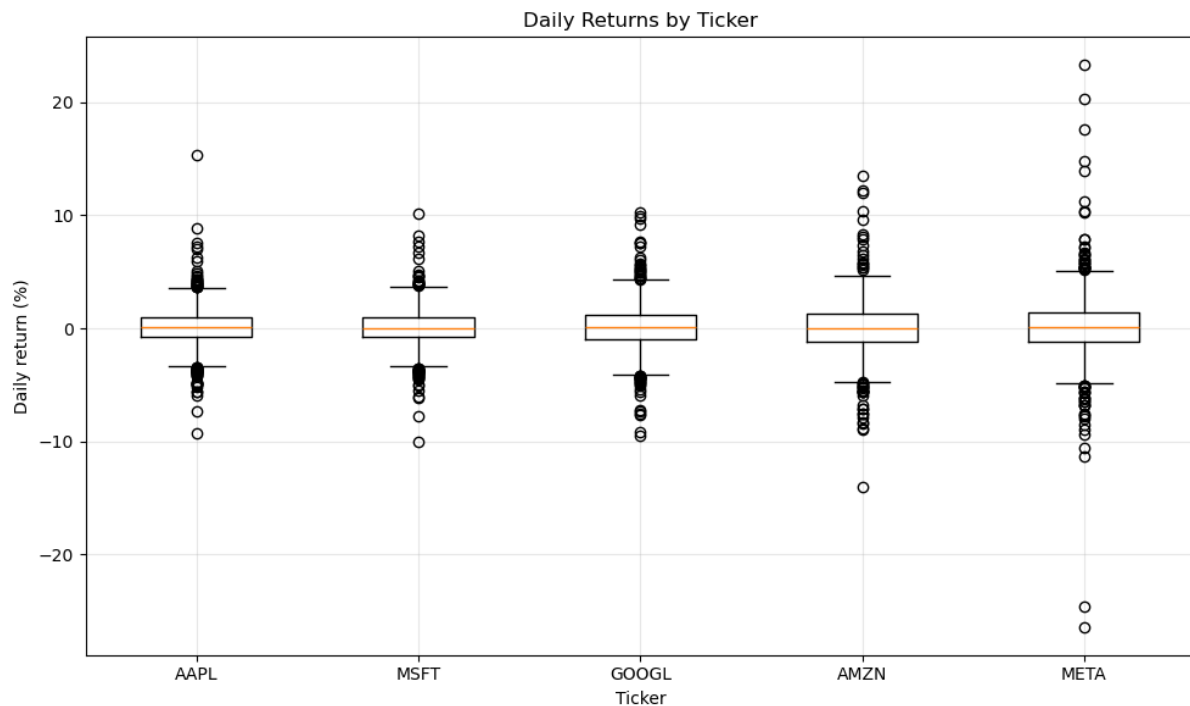


Figure 5: The box plot of daily returns of each ticker

To compare the central tendencies and support your ANOVA test, the bar chart was created with 95% of confidence intervals (Figure 6), representing every error bar crossing zero and overlapping all the others. This is the visual signature of a failure to reject the ANOVA.

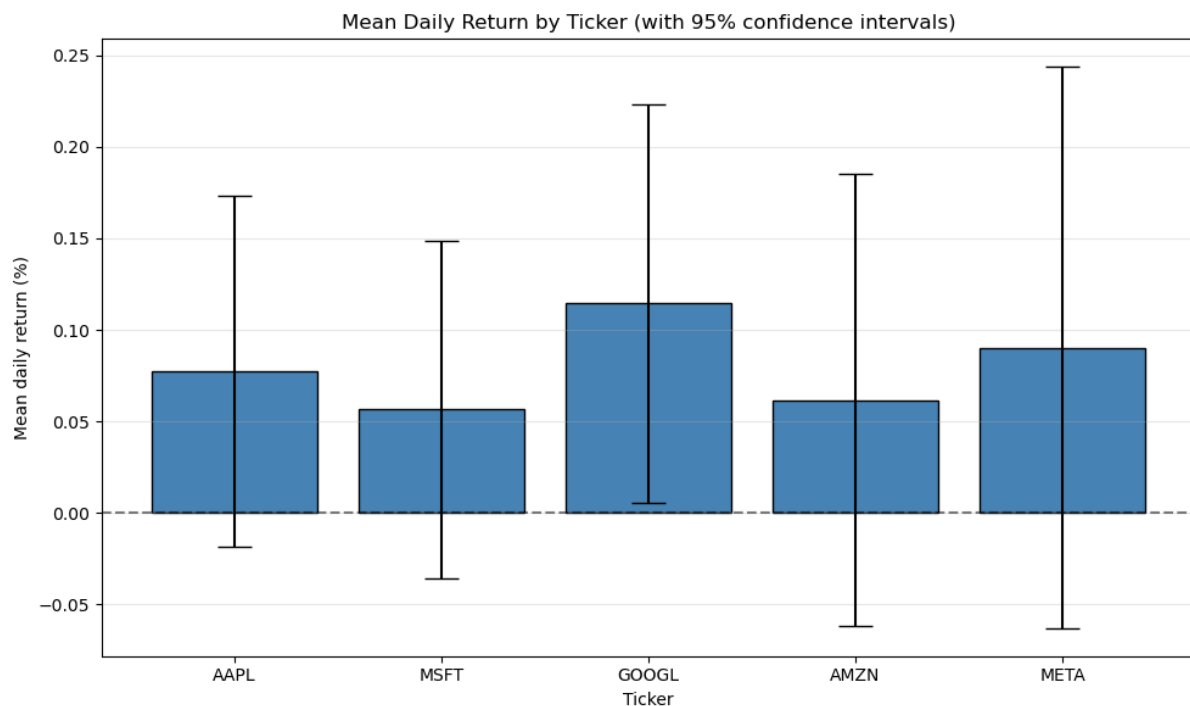


Figure 6: The Bar chart of mean daily returns with 95% confidence interval error bars

Conclusion

In real time, picking among the magnificent 5 each day is more like a coin toss than an investment skill for long-term performance. The differences are real, but they only emerge after compounding a huge number of days of statistically insignificant averages.

6.4 Insight 4 — Welch's t-test on log Trading Volume: Technology vs Finance

Question: Do Technology companies trade more actively than Financial companies on NASDAQ?

This insight is tested to know whether the technology companies trade more actively than the financial companies on NASDAQ. After loading the cleaned screener, the dataset was kept for only two sectors to compare. There are 769 technology and 1573 finance companies. And after checking the mean and median values, the right-skew was found. Therefore, the log-transformed np.log1p (Harris, C.R., Millman, K.J., van der Walt, S.J. et al., 2020) was used to get the normalized distribution. Welch's two-sample t-test was then run with `scipy.stats.ttest_ind(equal_var=False)`. To check the effect size, a custom `compute_cohens_d` function was created.

From printing summary statistics, the technology group got a log mean of volume 12.243 (std=2.606), compared with 9.951 (std = 2.941) for Finance. The difference of 2.29 log units was found, which is roughly ten times the daily trading volume on the raw

data for tech companies. The test result of $t = 19.15$ and an extremely small p-value is too much below the 0.05 threshold. Therefore, the null hypothesis of equal mean log volumes is firmly rejected. On the other hand, Cohen's d value 0.0808 is in the large band of effect size.

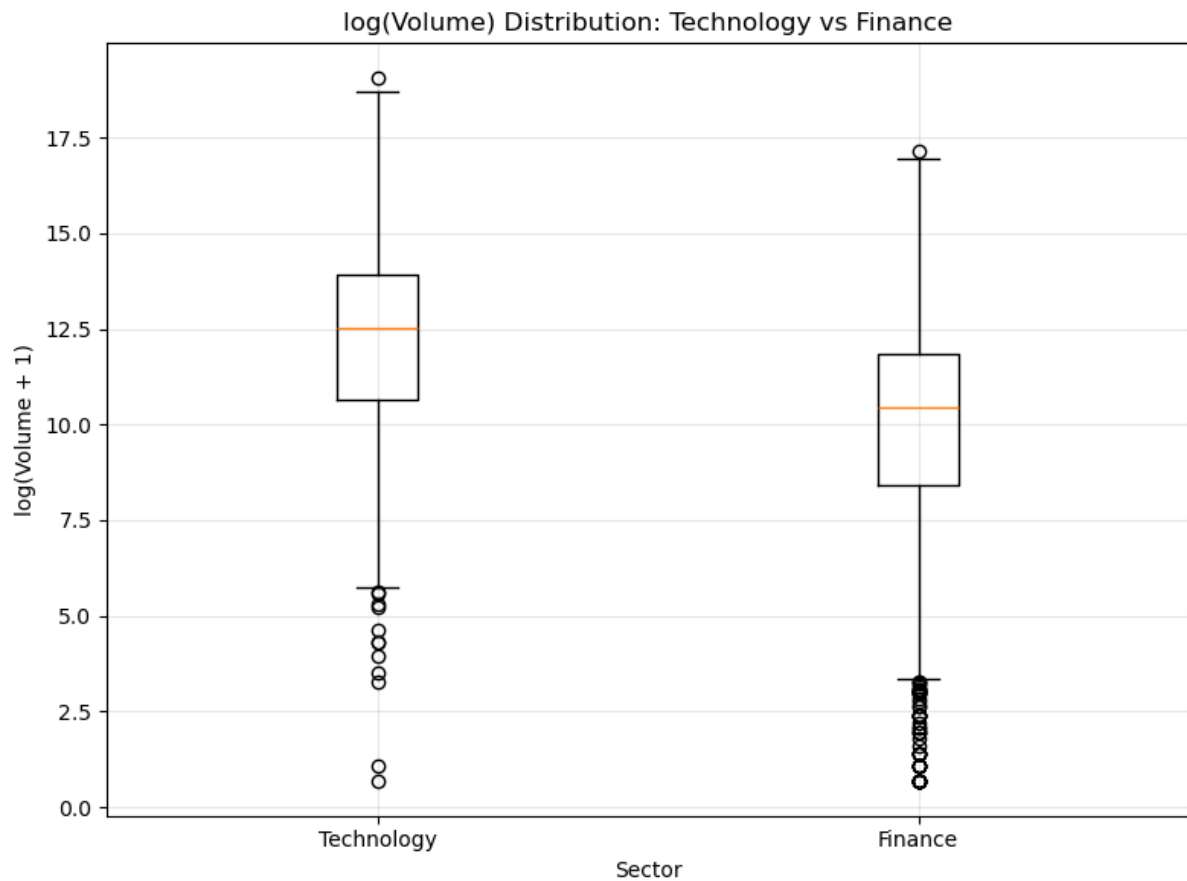


Figure 7: The box plot of log (Volume) distribution- Technology Vs Finance

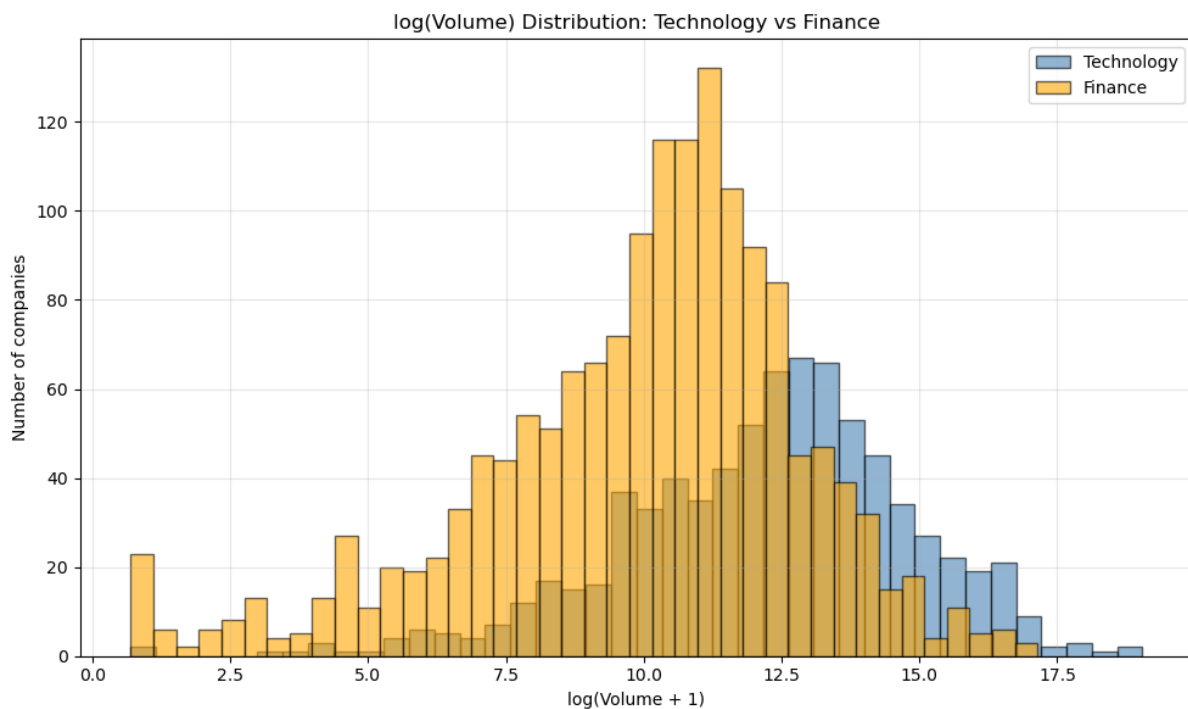


Figure 8: The overlapping histograms of log (Volume) distribution- Technology Vs Finance

The boxplot (Figure 7) shows the technology sector box sitting significantly above the financial sector box. The overlapping histograms (Figure 8) illustrate the two bell-shaped distributions clearly shifted from each other. This means that finance peaks around 10.5 log-volume, and technology does around 13.

Conclusion

In summary, the retail intuition in the stock market of “Tech is where the action is” is statistically supported with both an extremely small p-value and a large effect size. Another interesting point is that technology-sector companies on NASDAQ trade roughly ten times more actively than finance-sector ones on average.

7 Project Challenges

The project faced several practical challenges. When downloading the stock price data from Yahoo Finance, the downloaded dataset has a multi-level (Multiindex) column header, which had to be flattened first. After that, all ticker data could be merged into a single long-format DataFrame. Two columns in the NASDAQ screener, such as Last Sale and % Change) were as text values containing currency and percentage symbols, which were stripped using `str.replace` before the columns, and later changed to float type.

```
#Download daily data for each ticker in a loop, clean the columns, and collect the dataframes

frames = []
for ticker in tickers:
    data = yf.download(ticker, start=start_date, end=end_date, auto_adjust=True)
    #yfinance returns columns as a MultiIndex – made them to a single level
    if isinstance(data.columns, pd.MultiIndex):
        data.columns = data.columns.get_level_values(0)
    data["Ticker"] = ticker
    frames.append(data)
```

Another difficulty was the selection of statistical methods to derive the unique 4 insights from both datasets. An early idea considered a chi-square test on the sector versus the industry. Later, this was rejected because of the nested structure within the sectors, and it would produce meaningless results. The chi-square test was redesigned with the Market Cap tier instead in the Insight 2.

8 Conclusion

Overall, the four insights together give a clearer oversight of the Magnificent 5. What stands out from this project is that they differ a lot in return, volatility, and size. Therefore, they shouldn't be treated as a single investment. Across the wider NASDAQ market, the Technology sector belongs to the mega-caps, whereas Finance and Health Care contain mostly small companies. However, at the level of daily return, the Magnificent 5 are statistically the same as each other. On the other hand, Magnificent 5 shine in some places but not in others, so the real story is more complicated than the standard view..

9 References

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